```
"""Churn & retention analysis: RavenStack (SaaS) + Telco. Run: python analysis.py ->
dashboard.html"""
import pandas as pd, json
U='/mnt/user-data/uploads/'
a=pd.read_csv(U+'ravenstack_accounts.csv',parse_dates=
['signup_date']);s=pd.read_csv(U+'ravenstack_subscriptions.csv',parse_dates=
['start_date','end_date'])
c=pd.read_csv(U+'ravenstack_churn_events.csv',parse_dates=
['churn_date']);t=pd.read_csv(U+'ravenstack_support_tickets.csv')
d=pd.read_csv(U+'WA_Fn-UseC_-Telco-Customer-Churn.csv');d['ch']=(d.Churn=='Yes')
r=lambda x:round(float(x)*100,1)
def seg(df,col,flag): g=df.groupby(col)[flag].agg(['mean','count']);return
[[str(k),r(v['mean']),int(v['count'])] for k,v in
g.sort_values('mean',ascending=False).iterrows()]
mrr=s.groupby('account_id').mrr_amount.sum();a['mrr']=a.account_id.map(mrr).fillna(0)
a['q']=a.signup_date.dt.to_period('Q').astype(str)
R={}
R['rs_overall']=r(a.churn_flag.mean());R['rs_n']=len(a);R['rs_mrr_lost']=int(a[a.chur
n_flag].mrr.sum());R['rs_mrr']=int(a.mrr.sum())
R['rs_ind']=seg(a,'industry','churn_flag');R['rs_ref']=seg(a,'referral_source','churn
_flag');R['rs_ctry']=seg(a,'country','churn_flag');R['rs_plan']=seg(a,'plan_tier','ch
urn_flag')
R['rs_cohort']=[[k,r(1-v.mean()),len(v)] for k,v in a.groupby('q').churn_flag]
R['rs_reason']=[[k,int(v)] for k,v in c.reason_code.value_counts().items()]
R['rs_react']=int(c.is_reactivation.sum());R['rs_refund']=round(float(c.refund_amount
_usd.sum()))
tt=t.groupby('account_id').agg(n=('ticket_id','count'),sat=
('satisfaction_score','mean')).join(a.set_index('account_id').churn_flag)
R['rs_support']=[[('Churned' if k else
'Active'),round(v.n.mean(),2),round(v.sat.mean(),2)] for k,v in
tt.groupby('churn_flag')]
R['tc_overall']=r(d.ch.mean());R['tc_n']=len(d)
d['tenure_band']=pd.cut(d.tenure,[-1,12,24,48,72],labels=['0-12m','13-24m','25-
48m','49-72m'])
for k,col in [('contract','Contract'),('internet','InternetService'),
('pay','PaymentMethod'),('tech','TechSupport'),('tenure','tenure_band'),
('senior','SeniorCitizen')]: R['tc_'+k]=seg(d,col,'ch')
d['MonthlyCharges']=d.MonthlyCharges.astype(float);R['tc_rev_lost']=int(d[d.ch].Month
lyCharges.sum());R['tc_rev']=int(d.MonthlyCharges.sum())
json.dump(R,open('results.json','w'),indent=1)
html=open('/tmp/template.html').read().replace('__DATA__',json.dumps(R))
open('dashboard.html','w').write(html)
```